

PC-03 · EMULSIFIERS & THICKENERS · PERSONAL CARE

HS PEG-150 Distearate

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INCI	CAS	EC	FAMILY	SUPPLIED AS
PEG-150 Distearate	9005-08-7	—	PC-03	Per grade

01 POSITIONING

The HLB toolbox, supplied by grade — specified, verified, repeated.

02 HOW IT WORKS

- 01 HLB engineering
EO count sets the hydrophilic-lipophilic balance: each grade is a point on the HLB scale, combined to match the oil phase.
- 02 Interfacial film
Adsorbs at the oil–water boundary, dropping interfacial tension and stabilizing droplets against coalescence.
- 03 Electrolyte robustness
Non-ionic character keeps performance in hard water and salt-rich systems.

03 APPLICATIONS

- Creams and lotions (O/W)
- Fragrance solubilization in toners and micellar waters
- Bath oils
- Household and I&I emulsions
- Pharma-adjacent excipients (per grade)

04 FORMULATION GUIDE

USE LEVEL	0.5–6%
PH & CONDITIONS	Stable pH 3–10
— Match emulsifier pair HLB to the required HLB of the oil phase — the classic low+high combination beats any single grade.	
— For solubilization, premix solubilizer and fragrance 3:1 to 5:1 before adding to water.	

05 TYPICAL SPECIFICATION

Appearance	per grade: liquid / paste / flakes
HLB	per grade
Water content	≤ 1–3%
1,4-Dioxane	≤ 10 ppm (per TDS)

Typical parameters. The specification on the current TDS and the per-lot Certificate of Analysis govern.

06 PERFORMANCE — DIRECTIONAL

Emulsion stability (45 °C, class data) Paired HLB system	
Fragrance solubilization capacity PEG-40 HCO systems	

Values are representative of the ingredient class, based on published technical literature, and shown directionally for comparison. Final performance must be validated in the customer's formulation.